

Mathematical Modeling of The Black Death

Abstract

The world's most deadly pandemic to ever happen, the Black Death, killing over 100 million people and devastating Europe's population. While the timeline of the Black Death is scattered, evidence of the plague outbreak is seen in central Eurasian cemeteries in the early 1300s (Cohn SK Jr.). In 1346, there was evidence of major outbreaks around the Black Sea region. In 1347-1353, plague spread throughout Europe, the Middle East, and North Africa. The Black Death lasted nearly 500 years, killing 75-200 million people, decimating Europe's population, and killing up to 60% of the population in some cities (Cohn SK Jr.).

The plague originated from a bacterium called *Yersinia Pestis*, which originated near the Tian Shan mountains (Kyrgyzstan). In 1338, evidence of the bacteria was found in two cemeteries with tombstones labeled "pestilence," meaning fatal epidemic diseases (Spyrou, Musralina, and Gneccchi). The DNA sequences from the buried individuals confirmed infection by *Yersinia Pestis*. The transmission pathway is believed to be fleas, rodents, and humans. Humans were infected through flea bites or respiratory spread (Spyrou, Musralina, and Gneccchi). The plague lasted so long due to the differing strains that evolved from this bacterium, causing smallpox, scarlet fever, etc. All of these illnesses resulted from the same bacteria but evolved in different ways. The bacteria spread fast, due to environmental/ecological factors and the role of trade/human networks.

Climate fluctuations and ecological disruptions are environmental attributions that triggered the bacteria's host population change to rodents, causing the subsequent spread to humans. This is a result of altering the ecology of disease carriers, increasing the plague outbreak. Settlements that were near major trans-Asia trade routes are highly connected as they

linked central Asia with the Mediterranean and Europe (Galanaud, Galanaud, and Giraudoux).

The consequence is heavy traffic of people coming in and out. This movement of goods, animals, and people helped carry and spread the infected fleas and rodents. This is exactly how the plague was able to cover the entire continent. These highly connected trade routes cause a trickle-down effect of infections. Starting at a port city, transported to a smaller city, all the way down to small villages, infecting everyone along the way.

Introduction

As epidemics emerge, analysts will seek to show the projected (for new illnesses) or actual (for historical illnesses) trajectory the disease's spread will have, commonly executed using an SIR model. An SIR model has five main parameters: the size of the susceptible population, the size of the infected population, the size of the recovered population, the transmission rate, and the recovery rate (Choisy). The model assumes that once someone becomes infected, they will move out of the susceptible pool and into the recovered population and cannot get reinfected. Put simply, SIR models investigate the trends and rates of spread of an infectious disease within a population. In practice, the model will use statistics and differential equations to predict the size of an infected population at any given time, and show how that population changes. This modeling takes the epidemiology of diseases to a more sophisticated level than pure data can. The SIR model is varied, and specific populations can be investigated by changing the beta term (transmission rate) (Choisy). In this paper, we sought to investigate how applying weighted edges and nodes, classic concepts in graph theory, modifies the transmission rate. As an example, we assign the nodes of a graph to be cities in Italy, and the edges connecting the nodes are interactions (significant trade cities, significant religious cities, ect.) The more interactions a city has, the more edges going to one node, the higher the rate of

spread should be in that city. Specifically, we can see that the plague devastated important trade cities like Venice much worse than a small isolated parish such as Pentedeilo. Partnered with network theory, there is a clear and detailed model of infectious spread and cause using the SIR model. After the first major plague outbreak wiped out most of Europe, Italy saw a massive population boom, doubling from 1400-1600 to ~14 million people. It also became highly urbanized as it transitioned from feudalism to capitalism with the increase of trade and textile production. 30 cities had a population greater than 10,000, the most in Europe at this time. Nobility was now based on wealth and public service rather than the family you're born into. Poor people still reside in dirty and compact communities, while nobles live in relatively spaced out but much cleaner conditions. In terms of societal positions, women (still with limited rights and freedoms) were seen as matriarchs of the household, expected to wait on the man of the house. Whereas men were seen more in public and political life, and are expected to participate in all aspects of public life and excel in their trade.

Methods

Data was sourced from a GitHub file (Lazzari) containing detailed death records from the parish of Eufimia, Italy, from the years 1630 and 1631. We initially translated from Italian to English and cleaned the raw data, excluding any irrelevant notes and names. The data was first cleaned in Excel and then imported into R as a CSV file. A subsequent clean was then applied in R to reformat dates as well as adding the following columns: a time index to count the days since the start of the plague, a plague flag, and separate ages into groups (infant, child, teen, young adult, and middle age). Using the cleaned data and SIR model, we performed multiple analysis tests on differing factors from the data, such as female, male, and age, to study the in-transmission rate. Graphs were created to test various factors, such as gender by plague

deaths, age by plague deaths, and types of deaths. These plots showed the prevalence of the plague within that year as well as possible social connections.

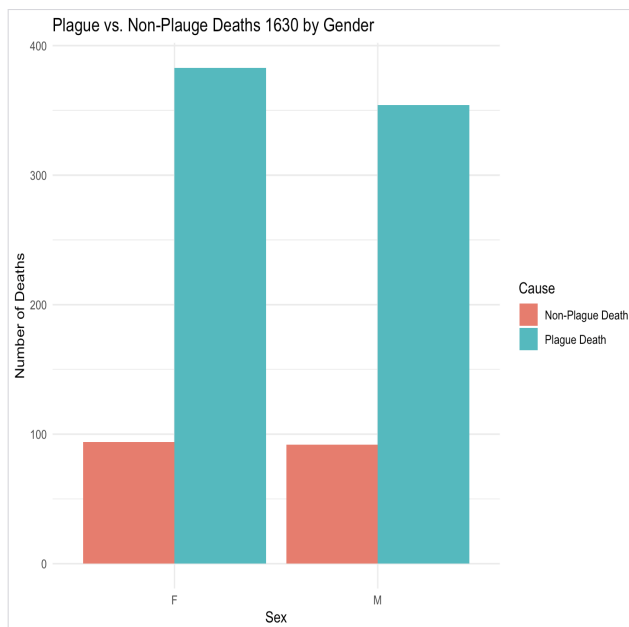


Figure 1. Plague vs. non-plague Deaths in 1630

From figure one, we can make a strong hypothesis that 1630 was either an initial or peak plague year for this parish. Concluded due to the extreme variation between plague and non-plague deaths. Conversely, we can see in figure two that the number of plague-related deaths has largely decreased relative to non-plague-related deaths.

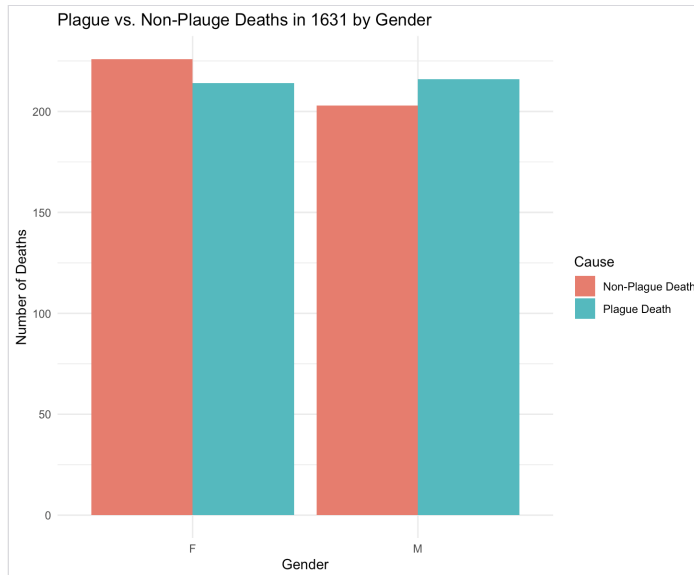


Figure 2. Plague vs. non-plague deaths in 1631

A possible conclusion for this is that the plague has killed most of the susceptible population, and there are few left to infect. The following graphs sought to investigate social correlations.

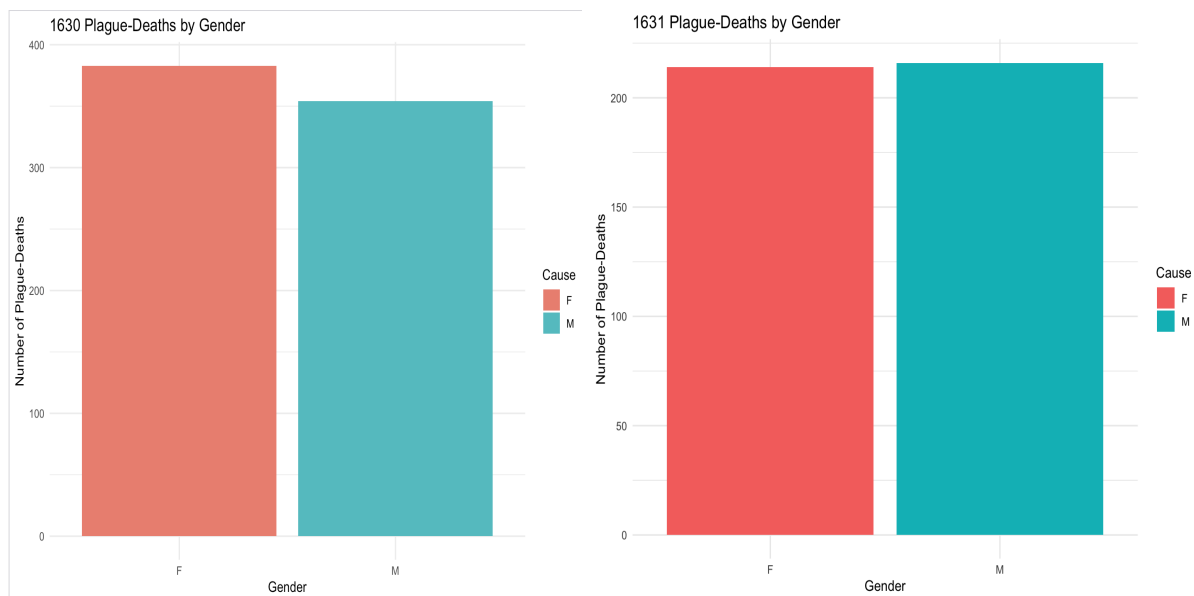


Figure 3. Plague deaths by gender in 1630/1631

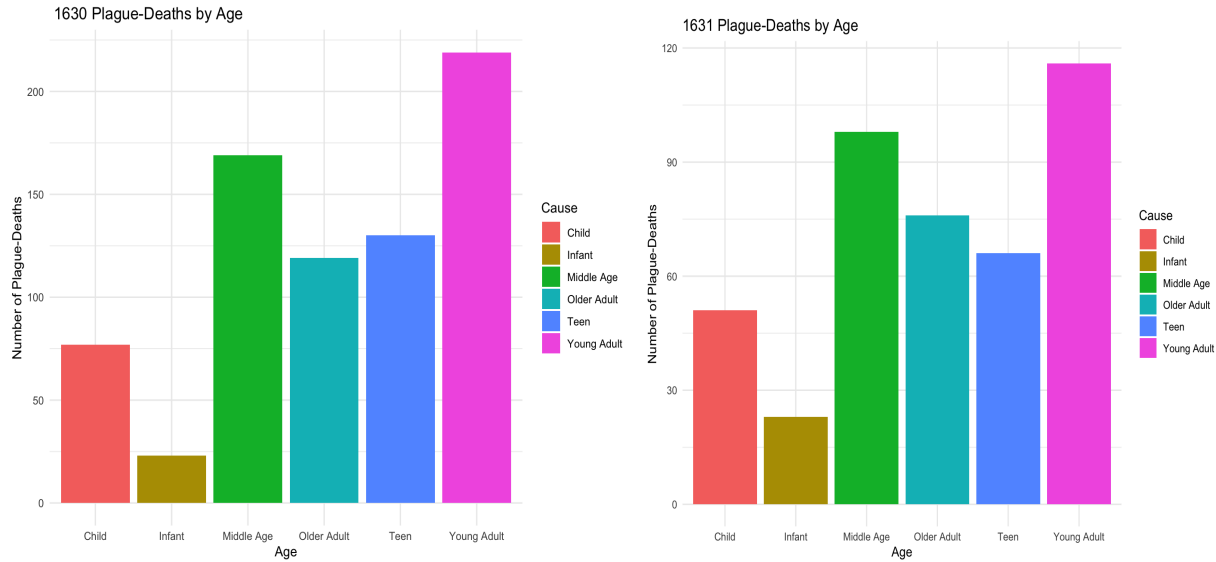


Figure 4. Plague deaths by age in 1630/31

The variation between the two years in figures 3 and 4 provides strong evidence that there were social implications that caused one group to be more heavily impacted than others.

To make sure our SIR model actually worked, we ran a generic test method using a beta value of 0.1625 and a gamma value of 0.125. These values were obtained by researching the average infectious period (8 days), where

$$\beta = 1.3 * \gamma$$

Where 1.3 is the average rate of spread.

$$\gamma = \frac{1}{\text{Average Infectious period}}$$

Finally, to confirm findings and for further analysis, we calculated the R_0 value for each test, where if $R_0 > 1$, we expect the disease to spread.

Data

In this section, we aim to show how societal norms and roles worked to increase the person-to-person contact within the spread of the plague. The general model that we will be using is shown in figure 5.

```
#-----NEW SIR MODEL-----##
# Things That wont Change
sir_model_f <- function(time, state, parameters) {
  with(as.list(c(state, parameters), plague_deaths_by_age_31 (gg, 81000 bytes))
    N <- S + I + R
    dS <- -beta * S * I / N
    dI <- beta * S * I / N - gamma * I
    dR <- gamma * I

    list(c(dS, dI, dR))
  })
}
alpha <- 2
beta_0 <- 0.1625
time_values_f <- seq(0, 350, by = 10) # Time span of Data in Days

## Parameters and Initial Value
parameters_values_f <- c(
  beta_f <- beta_net_f, # This is where Beta will differ from original model
  gamma <- 0.125
)
initial_values_f <- c(
  S = N_f - 1, # Population of Susceptable Females in Venice At 1/03/1631
  I = 1, # Number of Females infected on 01/03/1631
  R = 0 # Number of Females Recovered/Dead on 01/03/1631
)

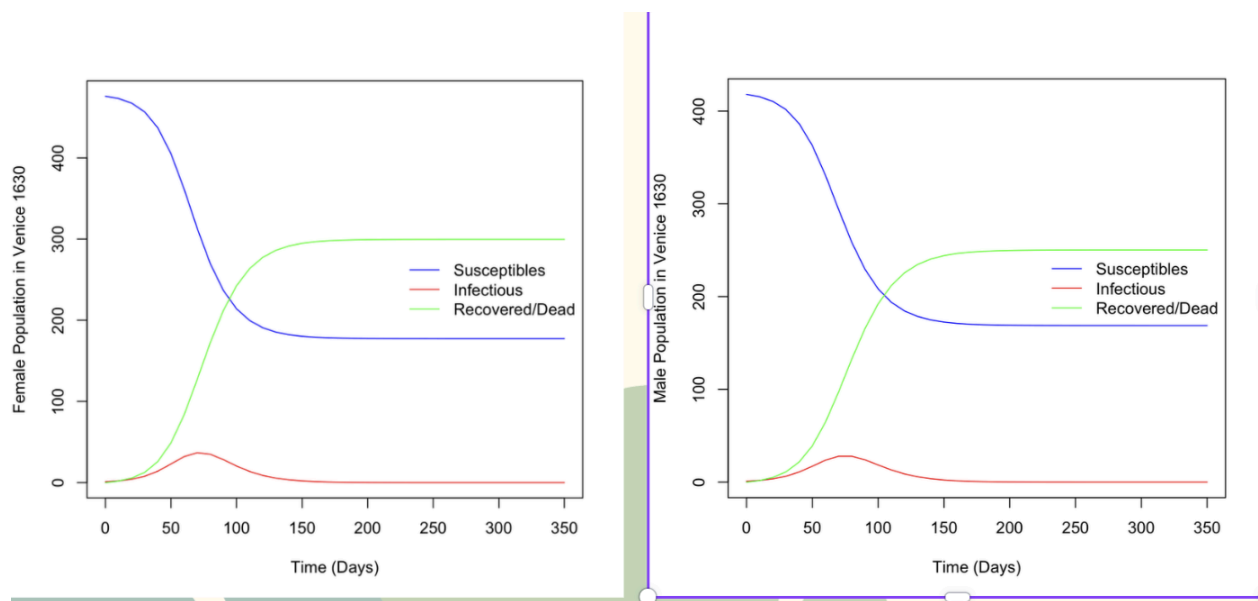
## SIR Modeling
sir_values_f <- ode(
  y = initial_values_f,
  times = time_values,
  func = sir_model_f,
  parms = parameters_values_f
)
sir_values_f <- as.data.frame(sir_values_f)

## Plotting Female's SIR Model
matplot(sir_values_f$time, sir_values_f[, c("S", "I", "R")],
  type = "l", lty = 1, col = c("blue", "red", "green"),
  xlab = "Time (Days)", ylab = "Female Population in Venice")
legend("right", c("Susceptibles", "Infectious", "Recovered/Dead"),
  col = c("blue", "red", "green"), lty = 1, bty = "n")
```

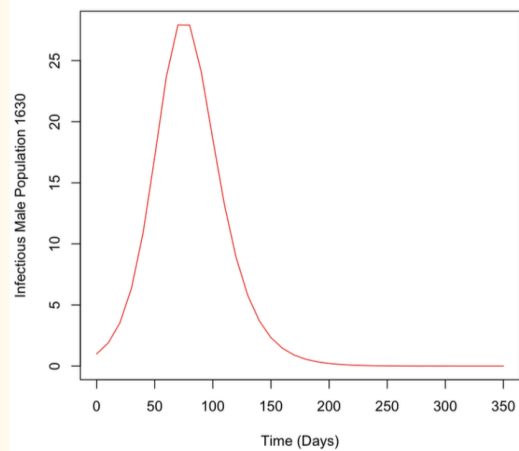
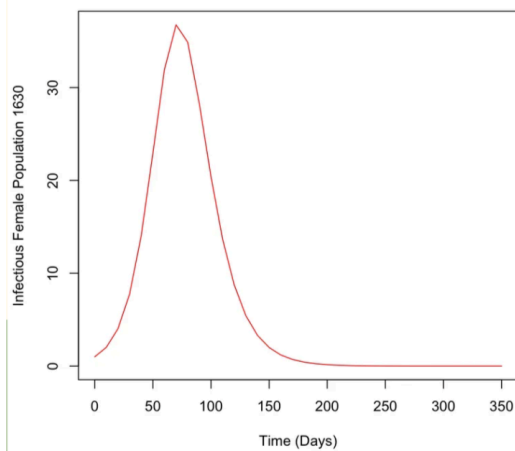
Figure 5. R model that will be varied

Using this model, for the year 1630, we will show how the rate of spread was different between females, males, and age groups. When working with this data, we have to make three assumptions. Firstly, we used the total amount of recorded deaths as our population size. This was necessary because there were no records on the population size of St. Eufemia. Since not all

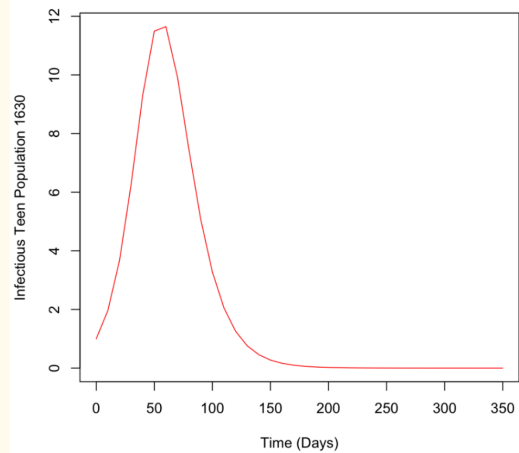
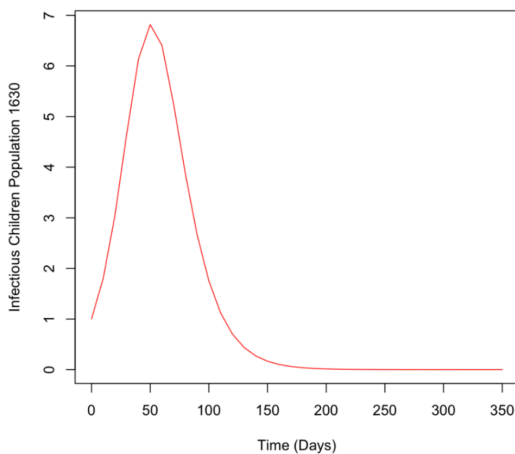
the deaths in these records are plague deaths, it still provides the variation that is seen when people are immune to a disease or don't get sick. Secondly, once a person is out of the infectious population and goes into the recovered/dead, they can not re-enter the susceptible population. This is just an inherent characteristic of SIR models. Finally, we are assuming that if two people are infected during the 8-day infectious window, this is a possible transmission path. We assign edges between people in the window; the more edges the person has, the larger their node weight. Now that our assumptions are established, we can go on to compare our different categories.



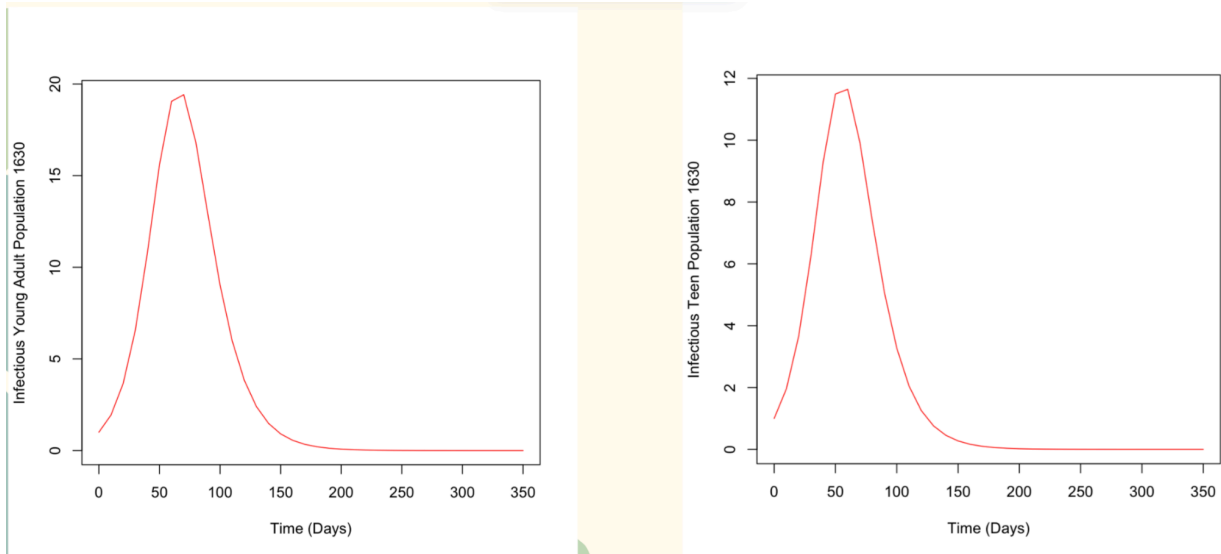
Comparing male and female models, we see that there is a difference between the peaks of the graph.



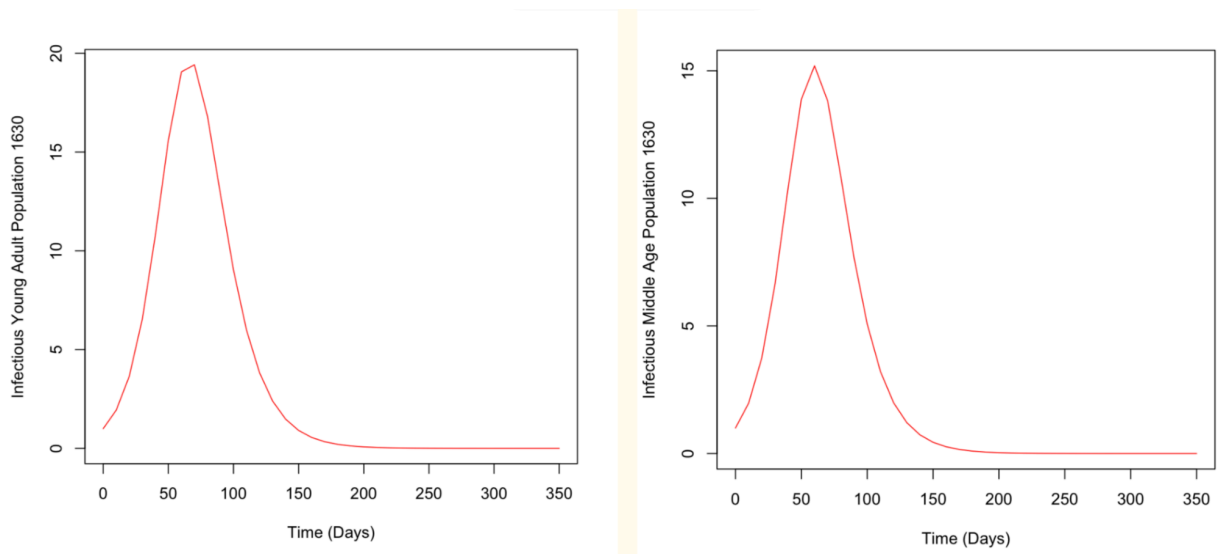
The flat plateau we see in the male graph is due to there being a period of time where the dying stopped increasing, but remained the same for a short period of time. However, the female graph increased fast and then decreased quickly too. Keeping in mind what the peaks and plateaus mean, let's take a look at comparing children vs. teen models, young adult vs. teen models, young adult vs. middle-aged models, and middle-aged vs. senior models.



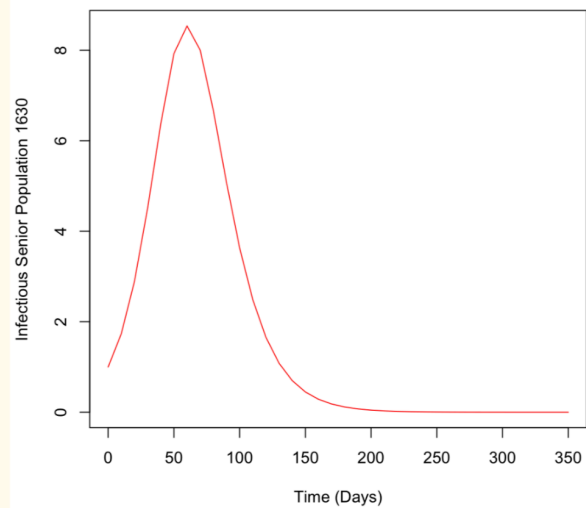
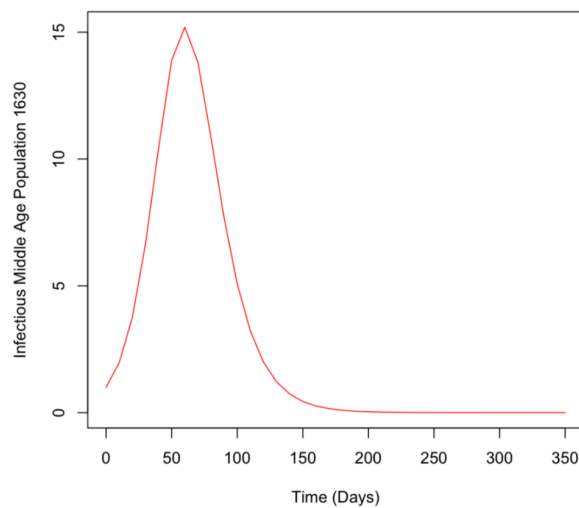
Children and Teen model



Young Adult and Teen Model



Young Adult and Middle Age Population



Middle Age and Senior Population

From these graphs, we can make the conclusion that there is variation between different populations.

Conclusion

The variation in the graphs could be attributed to the differences and similarities in the populations. For example, the Senior and middle-aged populations have a very similar distribution. This could be attributed to the fact that the life expectancy was much lower in this time period, plague aside. Similarly, young adults and teens would have nearly identical social roles given that they thought of 14-year-olds as adults. The difference between male and female populations is a direct consequence to the social expectations of the two groups. Women are more impacted because they have much higher rates of exposure through daily tasks. For example, they could be at risk of exposure from their family or markets when they do daily shopping. Finally, for further study, we would have liked to examine how to show this on a larger geographical scale to examine why the plague spread so far so fast through trade routes. If we

were able to find data on this, we think it would be very interesting to see the differences in isolated countries/cities and high traffic countries/cities.

Citations

Cohn SK Jr. Epidemiology of the Black Death and successive waves of plague. *Med Hist Suppl.* 2008;(27):74-100. PMID: 18575083; PMCID: PMC2630035.

Spyrou, M.A., Musralina, L., Gneccchi Ruscone, G.A. *et al.* The source of the Black Death in fourteenth-century central Eurasia. *Nature* 606, 718–724 (2022).
<https://doi.org/10.1038/s41586-022-04800-3>

Galanaud P, Galanaud A, Giraudoux P. Historical Epidemics Cartography Generated by Spatial Analysis: Mapping the Heterogeneity of Three Medieval "Plagues" in Dijon. *PLoS One.* 2015 Dec 1;10(12):e0143866. doi: 10.1371/journal.pone.0143866. PMID: 26625117; PMCID: PMC4666600

Lazzari, G., Colavizza, G., Bortoluzzi, F. *et al.* A digital reconstruction of the 1630–1631 large plague outbreak in Venice. *Sci Rep* 10, 17849 (2020).
<https://doi.org/10.1038/s41598-020-74775-6>
Choisy, Marc, et al. “*The SIR Model.*” GitHub Pages,
<https://choisy.github.io/DMo2019/sir.html>.